

SHUBHAM KUMAR

Machine Learning Engineer

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PROFESSIONAL SUMMARY

Machine Learning Engineer with hands-on experience building NLP pipelines, speech-processing systems, computer vision applications, and real-time fraud detection platforms. Skilled in Python, TensorFlow, PyTorch, XGBoost, MediaPipe, and Apache Kafka, with a strong foundation in predictive modeling and deep learning. Seeking an ML Engineer / Data Scientist internship or role to build and deploy scalable, production-grade AI solutions.

TECHNICAL SKILLS

- **Languages:** Python, SQL, C
- **ML/DL Frameworks:** TensorFlow, PyTorch, Keras, Scikit-learn, XGBoost, Pandas, NumPy
- **NLP & AI:** NLP, Transformers, Hugging Face, LLMs, Generative AI, Agentic AI, Computer Vision
- **Computer Vision Tools:** MediaPipe, OpenCV, CNN, LSTM
- **Visualization:** Matplotlib, Seaborn
- **Tools & Platforms:** Git, GitHub, Docker, Anaconda, Postman, FastAPI, Flask, Streamlit, Apache Kafka, REST APIs

PROJECTS

Real-Time Fraud Detection System | Python, XGBoost, Scikit-learn, Apache Kafka, FastAPI

- Built an end-to-end fraud detection platform processing live transaction streams via Apache Kafka, covering data preprocessing, feature engineering, model inference, and automated risk scoring.
- Applied transformer-inspired embedding techniques and ensemble learning (XGBoost) for behavioral pattern analysis, optimizing Precision, Recall, F1-score, and ROC-AUC.
- Designed a scalable microservice architecture with REST APIs (FastAPI) for low-latency, high-throughput fraud scoring, reducing manual investigation effort and false positives.

AI Sign Language Translator | Python, MediaPipe, CNN/LSTM, NLP, Streamlit, pytsx3/gTTS

- Built a real-time, end-to-end sign language translation system converting webcam hand gestures into spoken sentences through a modular computer-vision-to-speech pipeline.
- Implemented hand detection and landmark extraction using MediaPipe (21 key points per hand), with feature engineering and normalization for robust gesture representation.
- Trained a deep learning model (CNN/LSTM) for gesture classification with confidence filtering, then built an NLP word/sentence builder with grammar post-processing.
- Integrated speech synthesis (pytsx3 / gTTS) for text-to-audio conversion and deployed an interactive real-time UI using Streamlit for end users.

WORK EXPERIENCE

Python ML Intern — FinAnalyz Technologies Pvt. Ltd.

Jan 2026 – Apr 2026

- Researched and benchmarked Whisper (OpenAI) and T5-based decoder pipelines for regional-language coverage across 8 Indian languages — Hindi, Bengali, Tamil, Telugu, Kannada, Malayalam, Marathi, and Gujarati — for speech-to-text and text-to-speech applications.
- Engineered a vernacular voice bot supporting 8 regional Indian languages/dialects using SarvamAI and OpenAI Whisper for accurate speech-to-text, validating 3 API endpoints using Postman and Base64 encoding.
- Implemented and fine-tuned a T5 decoder model to optimize text-to-speech data pipelines, achieving ~85% recognition accuracy during testing, and explored duration-predictor hyperparameters to enable controllable speech rate.
- Collaborated with a 3-member team on Python-based financial analytics modules while maintaining strict adherence to data confidentiality and secure coding standards.

EDUCATION & CERTIFICATIONS

- **Bachelor of Technology, CSE (AI/ML)** — Bengal College of Engineering and Technology, MAKAUT, West Bengal | 2023 – 2027 Ongoing -> GPA(7.5)
- **Intermediate** — Mahavir College, Gaya Ji | 2022
- **Matriculation** — Jai Hind Public School, Vigyan Nagar, Bodhgaya | 2020
- **Complete Data Science, Machine Learning, Deep Learning & NLP Bootcamp** — Udemy, Completion Certificate